

# LOAN DATA ANALYST PROJECT

## Loan Application Approval & Risk Analysis

This portfolio project demonstrates a complete data-analyst workflow: data understanding, cleaning, feature engineering, exploratory analysis, SQL analysis, Python analysis, dashboard planning and business recommendations.

### Dataset & Scope

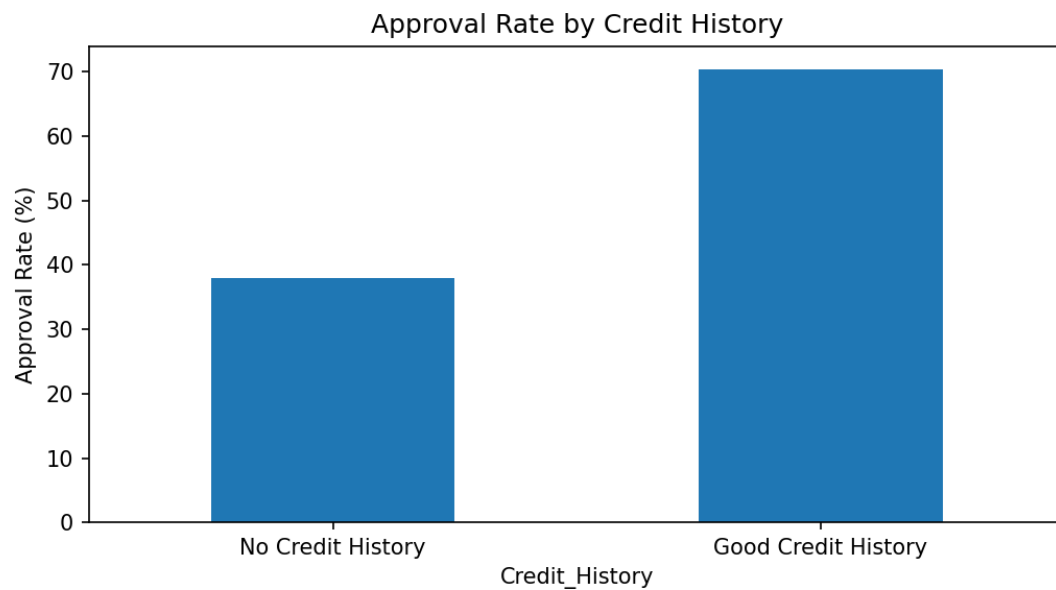
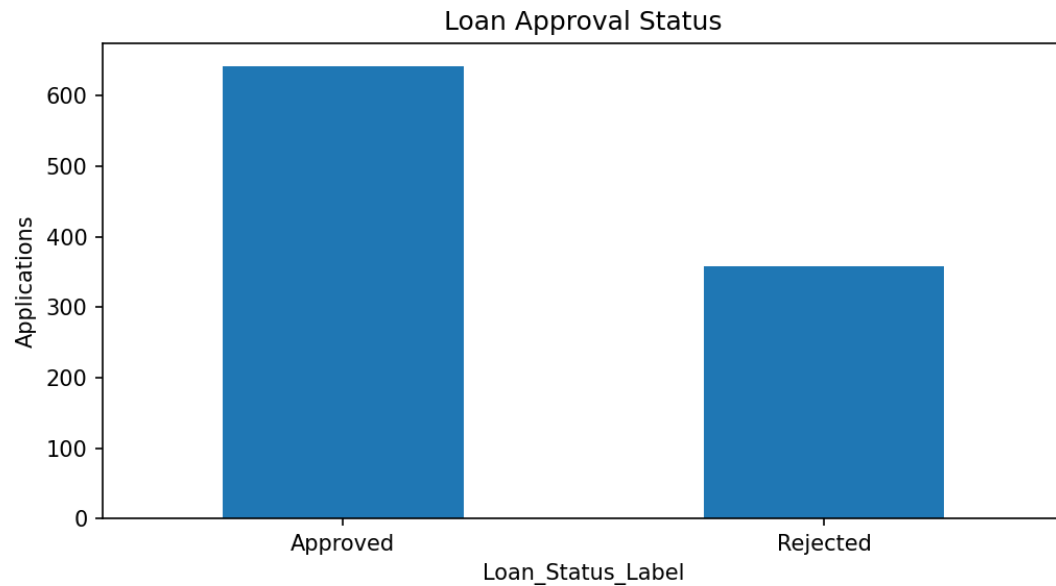
The included dataset is a realistic synthetic portfolio dataset following the structure of a commonly used public loan-approval dataset. It contains applicant demographics, income, loan details, credit history, property area and approval status. It is intended for portfolio/learning use.

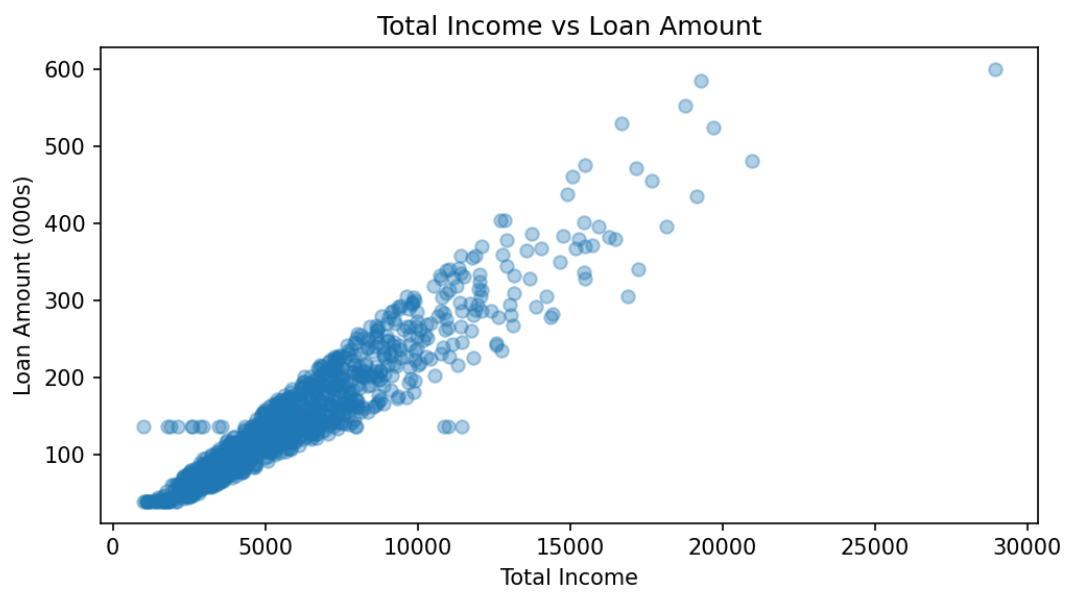
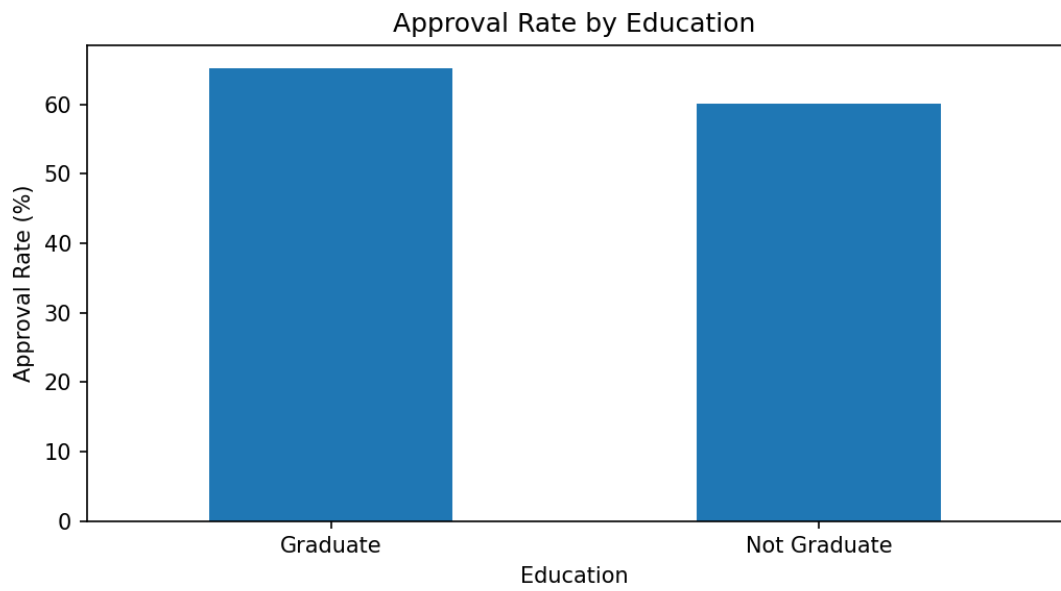
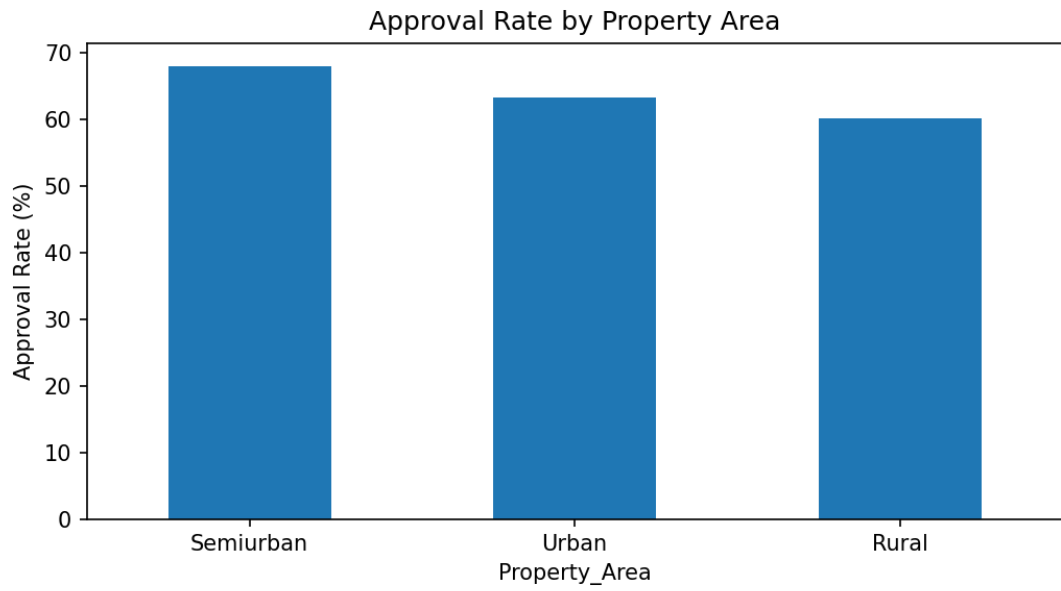
| KPI                  | Result       |
|----------------------|--------------|
| Total Applications   | 1,000        |
| Approved             | 642          |
| Rejected             | 358          |
| Approval Rate        | 64.20%       |
| Average Total Income | 6,197        |
| Average Loan Amount  | 155.2 (000s) |

## 1. Data Cleaning

Missing categorical values were filled using the mode. Loan amount was filled using the median, while loan term and credit history were filled using their modal values. The Dependents field was converted into a numeric helper field, and TotalIncome, Approval\_Flag, Loan\_to\_Income\_Ratio and Estimated\_EMI were created for analysis.

## 2. Exploratory Analysis





### 3. Key Findings

1. Credit history: the strongest screening signal in this dataset. The highest-credit-history segment has an approval rate of 70.4%.
2. Property area: Semiurban has the highest approval rate at 68.0%.
3. Education: Graduate applicants have the higher approval rate at 65.3%.
4. Income and loan amount should be assessed together because a large loan may be manageable for a high-income applicant but riskier for a low-income applicant.
5. Applications with poor credit history and above-average loan amounts should receive enhanced manual review.

### 4. SQL & Python Work

The SQL file contains KPI calculations, approval-rate analysis, segment comparisons, high-value loan queries and risk-screening queries. The Python script demonstrates loading the cleaned data, checking structure, and producing core approval analyses.

### 5. Power BI Dashboard Blueprint

| Page               | Recommended visuals                                                                                |
|--------------------|----------------------------------------------------------------------------------------------------|
| Executive Overview | Total Applications, Approval Rate, Approved/Rejected, Avg Loan Amount, Approval Trend/Distribution |
| Applicant Profile  | Gender, Marital Status, Education, Dependents, Self-Employment                                     |
| Credit & Risk      | Credit History vs Approval, Property Area vs Approval, Risk segment table                          |
| Income & Loan      | Income distribution, Loan Amount distribution, Income vs Loan Amount scatter                       |

### 6. Business Recommendations

- Prioritize credit-history verification early in the application process.
- Use income-to-loan ratio as a supplementary affordability indicator.
- Create a separate review queue for high-loan / weak-credit applications.
- Monitor approval rates by property area and applicant segment to identify operational differences.
- Use the Power BI dashboard as a management monitoring tool rather than as an automated approval decision.

### Portfolio Note

This project is designed to demonstrate analytical skills and should not be interpreted as a real bank underwriting model. A production lending model would require validated historical data, regulatory review, fairness testing, model governance and additional risk variables.